Software Requirements Specification (SRS)

Project Title: End-to-End Sales Analytics Dashboard: Revenue Trends, Product Performance, and Customer Segmentation

1. Introduction

1.1 Purpose & Scope

This document specifies the software and analytical requirements for a comprehensive retail/B2B sales data project. The system will process historical sales transactions to evaluate revenue trends, identify top-performing products, assess regional sales health, and segment the customer base. The project covers the complete data lifecycle: extraction, relational database management, exploratory data analysis (EDA), predictive analytics/segmentation, and executive dashboarding.

1.2 Key Objectives

- Calculate core financial metrics including Gross Margin, Year-over-Year (YoY) Growth, and Customer Lifetime Value (CLV).
- Perform Recency, Frequency, Monetary (RFM) analysis to segment customers into actionable marketing groups.
- Establish a robust data pipeline that moves from raw transactional logs to an interactive PowerBI dashboard tailored for Sales Directors and Regional Managers.

2. Data Requirements & Architecture

2.1 Data Sources & Extraction

The project relies on synthesizing transactional order data with customer and product catalogs:

- **Standard Dataset Location:** High-quality sales datasets are widely available on Kaggle. Excellent choices include the **"Global Superstore Dataset"** (for retail/B2B) or the **"Brazilian E-Commerce Public Dataset by Olist"** (for deep e-commerce sales).
- **Data Types:** * *Transactions:* Order_ID, Order_Date, Ship_Date, Customer_ID, Product_ID, Sales_Amount, Discount, Profit, Quantity.
    - *Product Catalog:* Product_ID, Category, Sub_Category, Product_Name, Base_Cost.
    - *Geography/Customer:* Customer_ID, Customer_Name, Segment (Consumer, Corporate), City, State, Region.

2.2 Data Preprocessing (Excel & Python)

Sales datasets often require cleaning to handle returns, missing geographic data, and date formatting.

- **Initial Screening (Excel):** Use Excel to spot-check the raw files. Apply conditional formatting to highlight negative Sales_Amount values (which typically indicate refunds/returns) to decide if they should be separated from gross revenue calculations.
- **Data Wrangling (Python / Pandas):**
    - **Datetime Indexing:** Convert Order_Date and Ship_Date into Pandas datetime objects to extract Year, Quarter, Month, and Day_of_Week.
    - **Handling Anomalies:** Identify and treat extreme outliers in the Quantity or Discount columns that could skew average order value calculations.
    - **Feature Creation:** Calculate the Net Profit Margin for each transaction directly in Pandas before loading it into the database.

2.3 Data Storage & Database Design (MySQL)

Cleaned transactional records will be stored in a relational database for structured, high-volume querying.

- **Schema Design:** Implement a classic Star Schema featuring a central Fact_Sales table connected to dimension tables: Dim_Customer, Dim_Product, Dim_Geography, and Dim_Time.
- **SQL Aggregations:**
    - Use GROUP BY and SUM() to calculate total revenue and profit by Product_Category and Region.
    - Utilize Window Functions (SUM() OVER(PARTITION BY Region ORDER BY Order_Date)) to calculate running Year-to-Date (YTD) totals for each specific sales territory.

3. System Features & Analytical Pipeline

3.1 Feature Engineering & Sales Metrics

Construct new variables using Python (NumPy, Pandas) to evaluate business performance:

- **Average Order Value (AOV):**
$$\text{AOV} = \frac{\text{Total Revenue}}{\text{Total Number of Orders}}$$
- **RFM Calculation:** Aggregate data at the Customer_ID level to calculate:
    - *Recency:* Days since the customer's last order.
    - *Frequency:* Total count of distinct orders.
    - *Monetary:* Total lifetime spend of the customer.

3.2 Exploratory Data Analysis (EDA)

Generate visual insights into sales behavior using Python (Matplotlib, Seaborn):

- **Pareto Charts (80/20 Rule):** Visualizing which 20% of products or customers are generating 80% of the total revenue or profit.
- **Time-Series Line Charts:** Plotting monthly sales revenue to visually demonstrate holiday seasonality or end-of-quarter spikes.
- **Correlation Heatmaps:** Mathematically mapping the relationship between Discount and Profit to prove at what discount threshold a product starts losing money.

3.3 Advanced Analytics (Machine Learning)

Utilize Python's scikit-learn to segment customers and forecast future sales:

- **Customer Segmentation (Unsupervised):**
  - *K-Means Clustering:* Feed the RFM scores into a K-Means algorithm to automatically group customers into distinct tiers (e.g., "Champions," "At-Risk," "Lost Cheap Customers"). Evaluate cluster quality using the Silhouette Score.
- **Sales Forecasting (Time-Series):**
  - *ARIMA or Prophet (Optional):* Train a time-series model on the aggregated weekly sales data to forecast the expected baseline revenue for the upcoming quarter, providing a target for the sales team.

4. Interactive Dashboarding (PowerBI)

The final deliverable is an operational PowerBI dashboard designed for the VP of Sales and territory managers.

- **Data Modeling:** Import the SQL schema and Python cluster outputs into PowerBI, establishing 1-to-Many relationships with the central Fact table.
- **DAX Calculations:** Create dynamic measures for YoY Revenue Growth %, Profit Margin %, and Total Customers.
- **Visual Interface:**
  - **Executive KPI Ribbon:** High-level cards displaying Total Sales, Total Profit, AOV, and Total Orders.
  - **Geospatial Sales Map:** A map visual plotting regions or cities, where bubble size represents Revenue and color saturation represents Profit Margin (highlighting regions that sell a lot but at a loss).
  - **Customer Segment Treemap:** Visualizing the proportion of the customer base that falls into each K-Means generated segment.
  - **Product Matrix:** A detailed table listing top products, conditionally formatted with data bars to show sales volume.
  - **Interactive Slicers:** Allow executives to instantly filter the dashboard by Year/Quarter, Region, and Product Category.

5. Project Delivery & Evaluation Structure

5.1 Directory Organization

The final project repository must be logically partitioned to facilitate technical review:

- /data: Raw transactional CSV extracts and the processed, segmented datasets.
- /scripts: Python modules (.ipynb) separated logically by workflow:
    - 1_data_cleaning_and_sql_formatting.ipynb
    - 2_eda_and_sales_metrics.ipynb
    - 3_rfm_calculation_and_kmeans_clustering.ipynb
- /sql: All DDL (table creation) and DML (aggregations, running totals, views for PowerBI) scripts.
- /reports: The finalized .pbix PowerBI dashboard file.

5.2 Code Quality & Maintainability

- **Documentation:** Python scripts must contain clear markdown cells explaining the business logic, especially regarding how returns (negative sales) were handled and the methodology for selecting the number of clusters (K) in the K-Means algorithm.
- **Data Integrity:** Strict enforcement of SQL Primary Keys (Order_ID) and Foreign Keys to ensure data consistency between the dimensional and fact tables, mimicking an enterprise data warehouse environment.